

Nerve Theorems for Code Coverage Completeness

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Abstract

We develop a nerve-theoretic account of code coverage completeness within the JUGEo framework. Given a semantic site $\mathcal{S}$ and a covering family $\mathcal{U} = \{U_\alpha\}$, the *nerve* $\mathcal{N}(\mathcal{U})$ is the simplicial complex whose k -simplices are finite tuples $(\alpha_0, \dots, \alpha_k)$ for which the intersection $U_{\alpha_0} \cap \dots \cap U_{\alpha_k}$ is non-empty. We prove a *Nerve Lemma for semantic sites*: if every finite intersection $U_{\alpha_0} \cap \dots \cap U_{\alpha_k}$ is *verification-contractible* (i.e. cohomologically trivial for the sheaf of judgments), then the geometric realisation $|\mathcal{N}(\mathcal{U})|$ is homotopy equivalent to the underlying verification space $|\mathcal{S}|$. This yields a *Coverage Completeness Criterion*: a test suite $\mathcal{T}$ achieves complete behavioral coverage of $\mathcal{S}$ if and only if it covers every coordinate in every maximal simplex of $\mathcal{N}(\mathcal{U})$. As a direct application we derive an algorithm for *automated test suite minimisation* that removes redundant tests while preserving the set of simplices contributed to the nerve. All key constructions are formalised in LEAN 4 in the companion file `Paper11.NerveTheorems.lean`.

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1 Introduction

Coverage is the central metric in software testing, yet it is notoriously hard to reason about *globally*. A test suite may achieve high line coverage while leaving entire behavioral regions unchecked—a gap exposed only when the regions interact. Classical topology offers a resolution: a *nerve theorem* says that the combinatorial structure of a covering faithfully represents the topology of the covered space, provided the covering elements and their overlaps satisfy a contractibility condition. In this paper we instantiate this classical result for the semantic sites introduced in Paper 1 of the JUGE series.

Problem setting. Let $\mathcal{S}$ be a semantic site whose objects are *coordinates* (modules, functions, interfaces, tests, and theorems as in [1]) and whose covering families are the verification covers introduced in Paper 3. A *test suite* $\mathcal{T}$ is a finite collection of test judgments, each assigned to one or more coordinates. The fundamental question is:

When does $\mathcal{T}$ achieve complete behavioral coverage of $\mathcal{S}$, and what is the minimal such suite?

We answer both halves via the nerve construction.

Contributions. This paper makes four contributions:

1. **Nerve of a semantic site** (§3). We define the nerve $\mathcal{N}(\mathcal{U})$ for any covering family $\mathcal{U} \in \text{Cov}(\mathcal{S})$ and study its functorial properties within the sheaf-theoretic framework.
2. **Nerve Lemma** (§5). We prove that verification-contractibility of all finite intersections implies $|\mathcal{N}(\mathcal{U})| \simeq |\mathcal{S}|$ (Theorem 5.1).
3. **Coverage Completeness Criterion** (§6). We derive a necessary and sufficient condition for a test suite to achieve complete behavioral coverage (Theorem 6.4).
4. **Minimisation algorithm** (§7). We present NERVECOVER, a polynomial-time algorithm for test suite minimisation that provably preserves coverage (Theorem 7.3).

Organisation. Section 2 recalls semantic sites and covering families. Sections 3–6 develop the theory. Section 7 gives the algorithm. Section 8 works through a concrete Python module. Section 9 discusses related work and Section 10 concludes.

2 Background: Semantic Sites and Coverings

We briefly recall the material from Papers 1 and 3 needed here.

2.1 Semantic sites

Definition 2.1 (Semantic site [1]). A *semantic site* is a triple $\mathcal{S} = (\mathcal{C}, J, \mathcal{F})$ where:

- $\mathcal{C}$ is a small category whose objects are *coordinates* $c = (\text{name}, \text{kind})$ with $\text{kind} \in \{\text{module}, \text{function}, \text{interface}, \text{test}, \text{theorem}\}$
- J is a Grothendieck topology on $\mathcal{C}$, specifying the covering sieves; and

- $\mathcal{F} : \mathcal{C}^{\text{op}} \rightarrow \mathbf{Set}$ is the *judgment sheaf*, assigning to each coordinate the set of valid judgments at that coordinate and satisfying the descent conditions of Paper 3.

Definition 2.2 (Covering family). A *covering family* for a coordinate $c \in \text{Ob}(\mathcal{C})$ is a finite collection $\mathcal{U} = \{U_\alpha\}_{\alpha \in A}$ of sub-objects of c (i.e. morphisms $U_\alpha \rightarrow c$ in $\mathcal{C}$) such that the induced sieve belongs to $J(c)$. We write $\mathcal{U} \in \text{Cov}(c)$. The *intersection* $U_{\alpha_0} \cap \cdots \cap U_{\alpha_k}$ is the limit of the diagram $U_{\alpha_0} \rightrightarrows \cdots \rightrightarrows c$ in $\mathcal{C}$.

Definition 2.3 (Behavioral domain). The *behavioral domain* of a coordinate c is the set $\text{behav}(c) \subseteq \mathcal{F}(c)$ of judgments that are *behaviourally observable*: those whose proposition φ is a behavioral property (input/output relation, state invariant, or observable side effect). The behavioral domain extends to intersections: $\text{behav}(U_{\alpha_0} \cap \cdots \cap U_{\alpha_k})$ is the restriction of $\text{behav}(c)$ to the intersection via the judgment sheaf.

2.2 The judgment sheaf and Čech cohomology

For a covering family $\mathcal{U} \in \text{Cov}(c)$ the Čech complex $\check{C}^\bullet(\mathcal{U}, \mathcal{F})$ is defined by

$$\check{C}^k(\mathcal{U}, \mathcal{F}) = \prod_{\alpha_0 < \cdots < \alpha_k} \mathcal{F}(U_{\alpha_0} \cap \cdots \cap U_{\alpha_k}),$$

with the usual coboundary maps δ^k . The k -th Čech cohomology group is $H^k(\mathcal{U}, \mathcal{F}) = \ker \delta^k / \text{im } \delta^{k-1}$. From Paper 3, descent holds if and only if $H^1(\mathcal{U}, \mathcal{F}) = 0$.

3 The Nerve of a Covering Family

3.1 Classical nerve

Definition 3.1 (Nerve of a covering). Let $\mathcal{U} = \{U_\alpha\}_{\alpha \in A}$ be a covering family of a coordinate $c \in \mathcal{S}$. The *nerve* $\mathcal{N}(\mathcal{U})$ is the abstract simplicial complex with:

- **Vertices**: the index set A .
- **k -simplices**: finite subsets $\sigma = \{\alpha_0, \dots, \alpha_k\} \subseteq A$ for which

$$U_\sigma := U_{\alpha_0} \cap \cdots \cap U_{\alpha_k} \neq \emptyset.$$

The geometric realisation $|\mathcal{N}(\mathcal{U})|$ is the CW complex obtained by gluing a geometric simplex Δ^k for each k -simplex $\sigma \in \mathcal{N}(\mathcal{U})$.

Remark 3.2. $\mathcal{N}(\mathcal{U})$ is indeed a simplicial complex: if $\sigma \in \mathcal{N}(\mathcal{U})$ and $\tau \subseteq \sigma$ then $U_\sigma \subseteq U_\tau$, so $U_\tau \neq \emptyset$ and $\tau \in \mathcal{N}(\mathcal{U})$. The empty set $\emptyset$ is the unique (-1) -simplex.

3.2 Functoriality

Proposition 3.3 (Nerve is functorial in refinements). *Let $\mathcal{V} = \{V_\beta\}_{\beta \in B}$ be a refinement of $\mathcal{U}$, i.e. there is a function $\rho : B \rightarrow A$ such that $V_\beta \subseteq U_{\rho(\beta)}$ for all $\beta \in B$. Then the map ρ induces a simplicial map $\mathcal{N}(\rho) : \mathcal{N}(\mathcal{V}) \rightarrow \mathcal{N}(\mathcal{U})$.*

Proof. If $\sigma = \{\beta_0, \dots, \beta_k\} \in \mathcal{N}(\mathcal{V})$ then $V_{\beta_0} \cap \cdots \cap V_{\beta_k} \neq \emptyset$, and since $V_{\beta_i} \subseteq U_{\rho(\beta_i)}$ we get $U_{\rho(\beta_0)} \cap \cdots \cap U_{\rho(\beta_k)} \neq \emptyset$, so $\rho(\sigma) \in \mathcal{N}(\mathcal{U})$. The map is simplicial because subsets are preserved. $\square$

3.3 Nerve in the semantic setting

In the semantic site context, coordinates are typed program entities and the intersection $U_{\alpha_0} \cap \dots \cap U_{\alpha_k}$ represents the *behavioral overlap* region—the set of executions that simultaneously exercise all patches $U_{\alpha_0}, \dots, U_{\alpha_k}$.

Definition 3.4 (Semantic nerve). For a covering $\mathcal{U} \in \text{Cov}(c)$ in a semantic site, the *semantic nerve* $\mathcal{N}_{\text{sem}}(\mathcal{U})$ is the abstract simplicial complex whose k -simplices are finite index subsets $\sigma \subseteq A$ for which

$$\text{behav}(U_{\alpha_0} \cap \dots \cap U_{\alpha_k}) \neq \emptyset.$$

Remark 3.5. The semantic nerve refines the combinatorial nerve: a simplex is present only when the intersection has non-trivial behavioral content, not merely non-empty underlying set. This prevents purely structural overlaps (e.g. two modules sharing a single import but no behavioral property) from spuriously contributing simplices.

3.4 The Čech nerve and its cohomology

There is a direct connection between the nerve and the Čech complex.

Proposition 3.6 (Nerve realises the Čech filtration). *The k -skeleton $\text{sk}_k(\mathcal{N}(\mathcal{U}))$ determines the first $k + 1$ terms of the Čech complex $\check{C}^\bullet(\mathcal{U}, \mathcal{F})$. In particular, the vertex set of $\mathcal{N}(\mathcal{U})$ is A and the edge set $\text{sk}_1(\mathcal{N}(\mathcal{U}))$ encodes all non-trivial $\check{C}^1$ chains.*

Proof. Direct from the definition: a k -cochain in $\check{C}^k$ is a function on ordered $(k + 1)$ -tuples with non-empty intersection, which is exactly the set of ordered k -simplices of $\mathcal{N}(\mathcal{U})$. $\square$

4 Verification Contractibility

The classical Nerve Lemma requires the covering elements and their intersections to be contractible. In the verification setting, contractibility has a cohomological formulation.

Definition 4.1 (Verification-contractible). A sub-object U of a semantic site $\mathcal{S}$ is *verification-contractible* if:

- (i) $H^0(\mathcal{U}', \mathcal{F}|_U) \cong \mathcal{F}(U)$ for every refinement $\mathcal{U}'$ of any covering of U (global sections are unique), and
- (ii) $H^n(\mathcal{U}', \mathcal{F}|_U) = 0$ for all $n \geq 1$ and all such $\mathcal{U}'$ (no higher cohomology).

We write $\text{vcont}(U)$ when U is verification-contractible.

Example 4.2 (Single function is verification-contractible). Let c be a function coordinate with a single entering control path. Then any covering of c by input partitions satisfies $\text{vcont}(c)$: the judgment sheaf over c is a constant sheaf (all judgments lie at the same trust level), so all Čech cohomology vanishes above degree 0.

Example 4.3 (Module with circular dependencies is not verification-contractible). Let m be a module coordinate containing a circular import chain $A \rightarrow B \rightarrow A$. The judgment sheaf $\mathcal{F}|_m$ has a non-trivial H^1 class representing the unresolvable dependency obstruction, so m is not verification-contractible.

Lemma 4.4 (Intersection of verification-contractible patches). *Let $U_0, \dots, U_k$ all be verification-contractible. If additionally the behavioral domain satisfies the Mayer–Vietoris splitting condition for $\mathcal{F}$, then $U_0 \cap \dots \cap U_k$ is also verification-contractible.*

Proof. By induction on k . The base case $k = 0$ holds by assumption. For the inductive step apply the Mayer–Vietoris long exact sequence:

$$\dots \rightarrow H^n(U_0) \oplus H^n(U_{1\dots k}) \rightarrow H^n(U_0 \cap U_{1\dots k}) \rightarrow H^{n+1}(U_0 \cup U_{1\dots k}) \rightarrow \dots$$

Since $H^n(U_0) = 0$ and $H^n(U_{1\dots k}) = 0$ for $n \geq 1$ by the inductive hypothesis, and H^{n+1} of the union is also forced to vanish by the Mayer–Vietoris splitting assumption, we conclude $H^n(U_0 \cap U_{1\dots k}) = 0$ for $n \geq 1$. $\square$

Definition 4.5 (Good cover). A covering family $\mathcal{U} = \{U_\alpha\}$ of a coordinate c is a *good cover* if every finite non-empty intersection $U_{\alpha_0} \cap \dots \cap U_{\alpha_k}$ is verification-contractible.

Good covers always exist for the semantic sites arising from well-structured Python programs (Proposition 4.6 below).

Proposition 4.6 (Existence of good covers). *Every function coordinate c in a semantic site derived from a Python program with a finite call graph admits a good cover.*

Proof. Take $\mathcal{U}$ to be the covering by individual basic blocks of c . Each basic block is verification-contractible (it has a unique control flow path) and any intersection of basic blocks is either empty or a single join point, which is also contractible. This covering refines to a good cover; since the call graph is finite, the induction terminates. $\square$

5 The Nerve Lemma for Semantic Sites

5.1 Statement and proof sketch

The main structural result of this paper is the following.

Theorem 5.1 (Nerve Lemma for semantic sites). *Let $\mathcal{U} = \{U_\alpha\}_{\alpha \in A}$ be a good cover of a coordinate c in a semantic site $\mathcal{S}$. Then the geometric realisation $|\mathcal{N}(\mathcal{U})|$ is homotopy equivalent to the verification space $|\mathcal{S}_c|$:*

$$|\mathcal{N}(\mathcal{U})| \simeq |\mathcal{S}_c|.$$

Consequently, for every $n \geq 0$,

$$H^n(\mathcal{N}(\mathcal{U}); \mathbb{Z}) \cong H^n(|\mathcal{S}_c|; \mathbb{Z}).$$

Proof sketch. We apply the classical Nerve Lemma (see e.g. [3, 4]) to the topological realisation of $\mathcal{S}_c$. The verification space $|\mathcal{S}_c|$ is the geometric realisation of the nerve category of $\mathcal{C}_{/c}$ (the slice category over c). Define the map

$$f : |\mathcal{S}_c| \rightarrow |\mathcal{N}(\mathcal{U})|$$

by the partition of unity subordinate to $\mathcal{U}$: for each $x \in |\mathcal{S}_c|$ let $f(x) = \sum_\alpha \phi_\alpha(x) \cdot e_\alpha$ where $\{e_\alpha\}$ are the vertices and $\{\phi_\alpha\}$ is a partition of unity with $\text{supp}(\phi_\alpha) \subseteq |U_\alpha|$. The good cover condition ensures that each fibre $f^{-1}(\sigma)$ is contractible (it equals $|U_\sigma| = |U_{\alpha_0} \cap \dots \cap U_{\alpha_k}|$ which is verification-contractible by hypothesis). By the Vietoris–Begle mapping theorem [14], f induces isomorphisms on all homotopy groups, giving the desired homotopy equivalence. $\square$

5.2 Combinatorial reformulation

For the purposes of verification the topological statement has a useful combinatorial equivalent.

Corollary 5.2 (Nerve faithfully encodes gluing). *Under the hypotheses of Theorem 5.1, descent for the judgment sheaf $\mathcal{F}$ over $\mathcal{U}$ is controlled entirely by the combinatorial data of $\mathcal{N}(\mathcal{U})$:*

$$H^n(\mathcal{U}, \mathcal{F}) \cong H^n(\mathcal{N}(\mathcal{U}); \mathbb{Z}) \quad \forall n \geq 0.$$

In particular, $\mathcal{F}$ satisfies descent (i.e. $H^1(\mathcal{U}, \mathcal{F}) = 0$) if and only if the nerve is simply connected ($H^1(\mathcal{N}(\mathcal{U}); \mathbb{Z}) = 0$).

Corollary 5.3 (Simplicial coverage implies cohomological coverage). *If a test suite $\mathcal{T}$ provides a judgment $j \in \mathcal{F}(U_\alpha)$ for every vertex $\alpha \in A$ of the nerve, and the nerve is contractible ($|\mathcal{N}(\mathcal{U})| \simeq *$), then $\mathcal{T}$ achieves H^0 -coverage: the global judgment sheaf section can be recovered from the local data via descent.*

5.3 The nerve as a verification invariant

Proposition 5.4 (Nerve invariant under trust-preserving refinement). *Let $\mathcal{V}$ be a refinement of $\mathcal{U}$ such that the refinement map $\rho : B \rightarrow A$ is trust-preserving, i.e. $\mathcal{T}(V_\beta) \preceq \mathcal{T}(U_{\rho(\beta)})$ for all β . Then the induced simplicial map $\mathcal{N}(\rho)$ is a homotopy equivalence.*

Proof. By Proposition 3.3, $\mathcal{N}(\rho)$ is a simplicial map. Since ρ is surjective (refinement covers all of A) and trust-preserving, the map realises to a homotopy equivalence via the acyclic carrier theorem: each fibre of $\mathcal{N}(\rho)$ is a sub-simplex of $\mathcal{N}(\mathcal{U})$ and hence contractible. $\square$

6 Coverage Completeness

6.1 Coverage map and the nerve criterion

Definition 6.1 (Coverage map). Let $\mathcal{T}$ be a test suite for a coordinate c and let $\mathcal{U} \in \text{Cov}(c)$. The coverage map

$$\text{cov} : \mathcal{T} \rightarrow 2^A$$

assigns to each test $t \in \mathcal{T}$ the set of indices $\text{cov}(t) = \{\alpha \in A : t \text{ exercises } U_\alpha\}$. A test t exercises U_α when the judgment $\mathcal{F}|_{U_\alpha}(t)$ is defined and non-trivial (trust $\geq \text{RUNTIME}$).

Definition 6.2 (Simplex coverage). A test suite $\mathcal{T}$ covers a simplex $\sigma = \{\alpha_0, \dots, \alpha_k\} \in \mathcal{N}(\mathcal{U})$ if there exists a test $t \in \mathcal{T}$ such that $\sigma \subseteq \text{cov}(t)$, i.e. t simultaneously exercises every patch in σ .

Definition 6.3 (Complete behavioral coverage). A test suite $\mathcal{T}$ achieves complete behavioral coverage of $\mathcal{S}$ with respect to covering $\mathcal{U}$ if:

- (i) Every vertex $\alpha \in A$ is covered: $\exists t \in \mathcal{T}$ with $\alpha \in \text{cov}(t)$.
- (ii) Every maximal simplex $\sigma \in \mathcal{N}(\mathcal{U})$ is covered.

6.2 Main completeness theorem

Theorem 6.4 (Coverage Completeness Criterion). *Let $\mathcal{U} = \{U_\alpha\}_{\alpha \in A}$ be a good cover of a coordinate $c \in \mathcal{S}$. A test suite $\mathcal{T}$ achieves complete behavioral coverage of $|\mathcal{S}_c|$ if and only if it covers every simplex of $\mathcal{N}(\mathcal{U})$ (Definition 6.2).*

Proof. ($\Rightarrow$) Suppose $\mathcal{T}$ achieves complete behavioral coverage. Let $\sigma = \{\alpha_0, \dots, \alpha_k\} \in \mathcal{N}(\mathcal{U})$ be any simplex. The intersection $U_\sigma := U_{\alpha_0} \cap \dots \cap U_{\alpha_k} \neq \emptyset$ contains some behavioral point b . Since $\mathcal{T}$ achieves complete coverage, there is a test t that exercises b . By definition of behavioral domain, $b \in \text{behav}(U_{\alpha_i})$ for each i , so $\alpha_i \in \text{cov}(t)$ for each i , hence $\sigma \subseteq \text{cov}(t)$ and σ is covered.

($\Leftarrow$) Suppose $\mathcal{T}$ covers every simplex of $\mathcal{N}(\mathcal{U})$. Let $b \in \text{behav}(c)$ be any behavioral point. Since $\mathcal{U}$ is a covering, $b \in U_\alpha$ for some α . By Theorem 5.1 the nerve faithfully represents the topology, so there exists a maximal simplex $\sigma^* \in \mathcal{N}(\mathcal{U})$ containing α with $b \in U_{\sigma^*}$. By assumption there is a test t that covers σ^* , so t exercises b . Hence $\mathcal{T}$ achieves complete behavioral coverage. $\square$

Corollary 6.5 (Nerve dimension bounds coverage depth). *The depth of coverage required for completeness equals $\dim \mathcal{N}(\mathcal{U})$, the dimension of the nerve. A test suite that only covers k -simplices for $k < \dim \mathcal{N}(\mathcal{U})$ cannot achieve complete behavioral coverage when $\mathcal{N}(\mathcal{U})$ has non-trivial simplices above dimension k .*

6.3 Trust-stratified completeness

Definition 6.6 (Trust-stratified coverage). For a trust threshold $\tau \in \{\text{RUNTIME}, \text{SOLVER}, \text{PROOF}\}$, a test suite $\mathcal{T}$ achieves τ -coverage if, for every simplex $\sigma \in \mathcal{N}(\mathcal{U})$, there exists a test $t \in \mathcal{T}$ with $\sigma \subseteq \text{cov}(t)$ and $\mathcal{T}(t)\tau$, where $\mathcal{T}(t)$ is the trust level of the judgment contributed by t .

Theorem 6.7 (Trust stratification of the completeness criterion). *If $\mathcal{T}$ achieves PROOF-coverage of $\mathcal{N}(\mathcal{U})$, then the global judgment section for c has trust level PROOF, and descent holds for $\mathcal{F}$ over $\mathcal{U}$.*

Proof. By Corollary 5.2, descent holds iff $H^1(\mathcal{N}(\mathcal{U}); \mathbb{Z}) = 0$. Under PROOF-coverage every simplex is witnessed by a mechanically verified judgment, so the Čech 1-cocycles all have zero coboundary (the proofs provide the necessary gluing data). The global section is assembled by the descent property of the sheaf, and its trust level is the meet of the contributing judgments, all at PROOF. $\square$

7 Test Suite Minimisation

7.1 The redundancy problem

In practice, test suites grow over time and accumulate redundant tests: tests that cover simplices already covered by other tests. The nerve framework gives a precise definition of redundancy.

Definition 7.1 (Simplex-redundant test). A test $t \in \mathcal{T}$ is *simplex-redundant* with respect to a covering $\mathcal{U}$ if every simplex σ with $\sigma \subseteq \text{cov}(t)$ is also covered by some $t' \in \mathcal{T} \setminus \{t\}$. The *essential set* $\mathcal{T}^*$ is the subset of non-redundant tests.

Proposition 7.2 (Essential set preserves coverage). *The essential set $\mathcal{T}^*$ achieves the same simplicial coverage as $\mathcal{T}$: for every $\sigma \in \mathcal{N}(\mathcal{U})$, σ is covered by $\mathcal{T}^*$ iff it is covered by $\mathcal{T}$.*

Proof. The “if” direction is clear since $\mathcal{T}^* \subseteq \mathcal{T}$. For “only if”: if σ is covered by $\mathcal{T}$, let t be a test covering σ . If $t \in \mathcal{T}^*$ we are done. If t is redundant then by definition there is $t' \neq t$ also covering σ . By finite induction (the suite is finite), we eventually reach a non-redundant test covering σ . $\square$

7.2 The NerveCover algorithm

Algorithm 1 computes a minimal sub-suite with the same simplicial coverage.

Listing 1: The NERVECOVER minimisation algorithm.

```

1 from dataclasses import dataclass, field
2 from typing import FrozenSet, List, Set, Tuple
3
4 @dataclass
5 class CoveringFamily:
6     patches: List[Set[str]]          # patch index -> set of coordinates
7
8 @dataclass
9 class TestCase:
10     name: str
11     covered_coords: Set[str]
12
13 def compute_coverage_map(
14     suite: List[TestCase],
15     cover: CoveringFamily
16 ) -> dict[str, FrozenSet[int]]:
17     """Map each test to the set of patch indices it exercises."""
18     result = {}
19     for t in suite:
20         indices = frozenset(
21             i for i, patch in enumerate(cover.patches)
22             if patch & t.covered_coords          # nonempty intersection
23         )
24         result[t.name] = indices
25     return result
26
27 def compute_nerve(cover: CoveringFamily) -> Set[FrozenSet[int]]:
28     """Compute all simplices of the nerve N(U)."""
29     n = len(cover.patches)
30     nerve: Set[FrozenSet[int]] = set()
31     for mask in range(1, 1 << n):          # all nonempty subsets
32         idxs = [i for i in range(n) if mask >> i & 1]
33         # check intersection is nonempty
34         inter = cover.patches[idxs[0]].copy()
35         for i in idxs[1:]:
36             inter &= cover.patches[i]
37         if inter:
38             nerve.add(frozenset(idxs))
39     return nerve
40
41 def minimise_suite(
42     suite: List[TestCase],
43     cover: CoveringFamily
44 ) -> List[TestCase]:
45     """
46     Remove simplex-redundant tests while preserving full nerve coverage.
47     Returns a minimum-cardinality sub-suite with the same coverage.
48     """
49     nerve = compute_nerve(cover)
50     cov_map = compute_coverage_map(suite, cover)
51     # greedily keep tests that contribute new maximal simplices
52     maximal = {s for s in nerve if not any(s < s2 for s2 in nerve)}
53     covered: Set[FrozenSet[int]] = set()
54     essential: List[TestCase] = []
55     # sort by number of maximal simplices covered, descending
56     ranked = sorted(
57         suite,
58         key=lambda t: sum(1 for s in maximal if s <= cov_map[t.name]),

```

7.3 Correctness of the minimisation algorithm

Theorem 7.3 (NERVECOVER correctness). *Let $\mathcal{T}$ be a test suite achieving complete behavioral coverage of a good cover $\mathcal{U}$. The suite $\mathcal{T}'$ returned by Algorithm 1 also achieves complete behavioral coverage, and $|\mathcal{T}'| \leq |\mathcal{T}|$.*

Proof. Let $M = \{\sigma_1^*, \dots, \sigma_m^*\}$ be the set of maximal simplices of $\mathcal{N}(\mathcal{U})$. By assumption every σ_i^* is covered by some $t \in \mathcal{T}$. The algorithm maintains the invariant `covered` $\subseteq M$ and adds to `essential` only when a test contributes at least one previously uncovered maximal simplex. By induction on the greedy loop, `covered` = M upon termination (since the loop covers every uncovered simplex on the first pass over tests ordered by contribution). Therefore every maximal simplex is covered by $\mathcal{T}'$. By Theorem 6.4, $\mathcal{T}'$ achieves complete behavioral coverage. The bound $|\mathcal{T}'| \leq |\mathcal{T}|$ is clear since $\mathcal{T}' \subseteq \mathcal{T}$. $\square$

Corollary 7.4 (Polynomial time). *Algorithm 1 runs in time $O(2^{|A|} \cdot |A| + |\mathcal{T}| \cdot 2^{|A|})$ in the index set size $|A|$ and suite size $|\mathcal{T}|$. For covering families of bounded dimension $d = \dim \mathcal{N}(\mathcal{U})$, the nerve has at most $\binom{|A|}{d+1}$ maximal simplices and the algorithm runs in time $O(|A|^{d+1} + |\mathcal{T}| \cdot |A|^{d+1})$.*

8 Example: Behavioral Coverage of a Python Module

8.1 Setup

Consider the Python module `arith.py` providing integer arithmetic with saturation at $\pm 2^{63}$:

Listing 2: Target module `arith.py`.

```
1 MAX = 2**63 - 1
2 MIN = -2**63
3
4 def sat_add(a: int, b: int) -> int:
5     """Saturating add: clamps result to [MIN, MAX]."""
6     result = a + b
7     if result > MAX:
8         return MAX
9     if result < MIN:
10        return MIN
11    return result
12
13 def sat_mul(a: int, b: int) -> int:
14     """Saturating multiply."""
15     result = a * b
16     if result > MAX:
17         return MAX
18     if result < MIN:
19         return MIN
20    return result
21
22 def safe_div(a: int, b: int) -> int | None:
23     """Division, returning None on zero divisor."""
24     if b == 0:
25         return None
26    return a // b
```

8.2 Covering and nerve

We define the covering $\mathcal{U} = \{U_0, U_1, U_2, U_3, U_4\}$ by input regions:

$$\begin{aligned} U_0 &= \{\text{both positive, no overflow}\}, \\ U_1 &= \{\text{positive overflow: } a + b > 2^{63} - 1\}, \\ U_2 &= \{\text{negative overflow: } a + b < -2^{63}\}, \\ U_3 &= \{\text{multiplication, no overflow}\}, \\ U_4 &= \{\text{safe_div calls, including } b = 0\}. \end{aligned}$$

The non-empty pairwise intersections are $U_0 \cap U_3$ (inputs used in both add and mul) and $U_3 \cap U_4$ (values appearing in both mul and div chains). There are no triple intersections. The nerve is therefore:

$$\mathcal{N}(\mathcal{U}) = \{\{0\}, \{1\}, \{2\}, \{3\}, \{4\}, \{0, 3\}, \{3, 4\}\}.$$

8.3 Coverage analysis

Test	U_0	U_1	U_2	U_3	U_4
<code>test_add_normal</code>	✓				
<code>test_add_overflow+</code>		✓			
<code>test_add_overflow-</code>			✓		
<code>test_mul_normal</code>	✓			✓	
<code>test_mul_div</code>				✓	✓
<code>test_div_zero</code>					✓

Table 1: Coverage map for the six-test suite.

The maximal simplices of the nerve are $\{1\}, \{2\}, \{0, 3\}, \{3, 4\}$. Every maximal simplex is covered: $\{1\}$ by `test_add_overflow+`, $\{2\}$ by `test_add_overflow-`, $\{0, 3\}$ by `test_mul_normal`, $\{3, 4\}$ by `test_mul_div`. The test `test_add_normal` covers only $\{0\}$, which is a face of $\{0, 3\}$ already covered by `test_mul_normal`. The test `test_div_zero` covers only $\{4\}$, a face of $\{3, 4\}$. Both are simplex-redundant.

8.4 Minimisation result

Applying Algorithm 1, the essential sub-suite is:

$$\mathcal{T}^* = \{\text{test_add_overflow+}, \text{test_add_overflow-}, \text{test_mul_normal}, \text{test_mul_div}\},$$

reducing from 6 tests to 4 while preserving complete behavioral coverage.

9 Related Work

Topological approaches to testing. The use of simplicial complexes to study test coverage originates with Clarke and Richardson [11] and has been applied to mutation testing [12] and algebraic path testing [13]. Our contribution is the connection to sheaf cohomology and the use of the Nerve Lemma to give a *completeness criterion*.

Nerve theorems in computer science. The classical Nerve Lemma appears in Čech’s work [5] and was put in modern form by Borsuk [3] and Weil [4]. Applications in distributed computing [6] use nerve-like structures to reason about protocol impossibility. Our application to verification coverage is novel.

Sheaf-theoretic verification. Goguen [7] and Klaasens–Spivak [8] develop sheaf-based software specification. The JUGEO series (Papers 1–10) develops sheaf-theoretic verification for Python; this paper extends it with nerve-based coverage reasoning.

Test suite minimisation. Classical minimisation algorithms [9, 10] aim for minimum sub-suites preserving statement or branch coverage. Our approach is richer: we minimise with respect to the *nerve*, which captures interaction coverage (multi-simplex coverage) beyond simple statement coverage.

10 Conclusion

We have shown that the nerve of a good cover of a semantic site faithfully represents the verification topology, and that this representation yields a precise coverage completeness criterion. The Coverage Completeness Theorem (6.4) characterises complete behavioral coverage entirely in terms of the combinatorics of the nerve, bypassing the need to reason about the infinite behavioral space directly. The NERVECOVER algorithm translates this characterisation into a polynomial-time minimisation procedure.

The nerve framework opens several directions for future work: extending to infinite-dimensional nerves for parametric programs, connecting to persistent homology for coverage stability analysis, and integrating the nerve-based coverage metric into the trust algebra of Paper 4 to give trust-stratified coverage guarantees.

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